

San Diego Movie Theater Ticketing

System Software Requirements Specification

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Group 1
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Prepared for
CS 250 — Introduction to Software Systems
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Revision History

Date	Version	Author	Comments
July 10, 2026	1.0	Anthony Alarcon et al.	Initial draft from client interviews and requirements elicitation

Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
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	Dr. Gus Hanna	Instructor, CS 250	

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to provide a complete, accurate and organized description of the requirements for the San Diego Movie Theater Ticketing System. It serves as the main structure for developers, testers, project managers, and other stakeholders during the development process. This document helps outline both the functional and non-functional requirements needed to create a secure, reliable, user-friendly, and a scaling ticketing system for a network of 20 movie theaters in San Diego County.

1.2 Scope

The SD Movie Theater Ticketing System is a web-based application that allows the customers to browse what movies are available and their showtimes, reserve seats, purchase up to 20 tickets in a single transaction, and receive either their tickets digitally or printed. Deluxe theaters will provide assigned seating, while regular theaters provide open seating. The system also offers the customers loyalty accounts, multiple payment options including credit/debit cards, PayPal, and Bitcoin, discounted pricing for eligible customers, post-purchase feedback, and access to movie ratings and reviews from sources such as Rotten Tomatoes and IMDb.

This system can easily be accessed through modern desktop and mobile web browsers, as well as kiosks that are located inside the theaters. It includes an administrative portal that enables theater staff to manage showtimes, process purchase overrides, and help resolve customer issues. To support business operations, the system keeps daily transaction logs, is designed to handle at least 1,000 customers that are operating it at the same time during peak hours, and uses security measures to generate unique, non-duplicable tickets that help prevent fraud and ticket scalping.

The initial release will not include native mobile applications, expanded locations outside San Diego County, subscription-based movie passes, or integration with third-party platforms.

1.3 Definitions, Acronyms, and Abbreviations

- **Actor:** A user or external system that interacts with the System in a particular role.
- **Deluxe Theater:** Theater with assigned seating and 75 seats.
- **Loyalty Account:** Optional registered customer profile storing personal information,

payment methods, purchase history, and loyalty points.

- **Regular Theater:** Theater with open seating and 150 seats.
- **Showing:** A specific movie screening at a particular date, time, and theater.
- **SRS:** Software Requirements Specification.
- **UC:** Use Case.
- **Unique Ticket:** A cryptographically secure or NFT-style ticket that cannot be duplicated or resold fraudulently.

1.4 References

- Client Interview Notes – Theater Ticketing System Qs.rtf (July 2026)
- Elicited Requirements – theater-ticketing-reqs.txt and theater-ticketing-questions.txt
- Additional Requirements – Theater Ticketing Requirements.rtf
- Lecture Notes: Scenarios and Use Cases – CS 5150, Cornell University (William Y. Arms)
- CS 250 SRS Template – Instructor Gus Hanna, Ph.D.

1.5 Overview

This SRS is split up into several sections. Section 2 explains the product, its users, and any limitations. Section 3 covers the system requirements, including the interfaces, functional requirements, use cases, and non-functional requirements. The remaining sections such as classes, analysis diagrams, and change management will be completed in later assignments.

2. General Description

2.1 Product Perspective

The SD Movie Theater Ticketing System is a web-based application that works with external services for payment processing (credit/debit cards, PayPal, and Bitcoin), movie reviews (Rotten Tomatoes and IMDb), and the theater's existing database. Customers can access the system through any web browser or kiosks that are located in the theater. The system also manages user accounts, loyalty information, and ticket purchases while keeping data updated across all theaters in the SD County area.

2.2 Product Functions

The primary functions of the System include:

- Browsing movies, showtimes, and theater information
- Seat selection and reservation (assigned in deluxe, open in regular)
- Ticket purchase (1–20 tickets) with multiple payment options and discounts
- Digital ticket delivery (email/app) and physical printing at kiosks/box office
- Optional loyalty account registration, login, and management
- Display of critic reviews and scores
- Post-purchase customer feedback collection
- Administrative functions for staff (showtime management, overrides, reporting)
- Security and anti-scalping measures (unique tickets,

single-device login)

2.3 User Characteristics

Main users of the system are moviegoers of all ages and experience levels who can buy tickets online or at kiosks. Some customers may have the option to choose to create loyalty accounts to receive additional benefits. Theater staff and managers are also users of the system and have access to administrative features for managing showtimes, handling customer issues, and viewing reports. The system is designed to be clear and manageable requiring little to no training.

2.4 General Constraints

- Must be delivered as a responsive web application (no native mobile app required for v1)
- Budget constraint: approximately \$500,000 for initial development
- Prototype ready for user testing within 4 months
- Limited to the San Diego metropolitan area (Pacific Time Zone) and 20 existing theaters
- Tickets valid only within the San Diego theater chain
- Minimal system downtime; updates must be performed with little to no service interruption

2.5 Assumptions and Dependencies

- Users have reliable internet access and a modern web browser.
- Payment gateways (credit card, PayPal, Bitcoin) are available and reliable.
- Existing theater showtimes and inventory databases can be integrated via API.
- Third-party review APIs (Rotten Tomatoes / IMDB) remain accessible.
- Theater staff are available for in-person refund handling and kiosk support.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The system will provide a user-friendly website that works well on desktop computers, mobile devices, and in-theater kiosks. The interface will support English, Spanish, and Swedish, and all prices will be displayed consistently using USD.

3.1.2 Hardware Interfaces

The system will work with standard web browsers on computers and mobile devices, as well as kiosks equipped with touchscreens, printers, and barcode/QR code scanners.

3.1.3 Software Interfaces

The system shall interface with:

- Payment gateway APIs (credit card processors, PayPal, Bitcoin)
- Movie review aggregation APIs (Rotten Tomatoes, IMDB)
- Existing partitioned theater showtime and inventory database

- Email/SMS gateway for digital ticket delivery

3.1.4 Communications Interfaces

The system will use secure HTTPS/TLS connections to protect all data sent between all users and the system. It will also constantly update seat availability and showtime information to ensure that everything is shown in real time so customers always see the most current information.

3.2 Functional Requirements

FR-01: Browse Showtimes and Movies

The system shall allow any user (guest or logged-in) to browse current and upcoming showtimes, movie details, and critic reviews/scores from Rotten Tomatoes and IMDB across all 20 San Diego theaters.

FR-02: Select Specific Seats

For deluxe theaters (75 seats), the system shall allow customers to view an interactive seat map and select specific seats. For regular theaters (150 seats), the system shall issue general admission tickets.

FR-03: Purchase Tickets (1–20 per Transaction)

Registered or guest users shall be able to purchase between 1 and 20 tickets for a single showing in one transaction. A 5-minute inactivity timer shall apply to the purchase alongside single session enforcement.

FR-04: Multiple Payment Methods

The system shall accept credit/debit cards, PayPal, and Bitcoin for ticket

purchases. **FR-05: Unique, Non-Replicable Tickets**

Every issued ticket shall be unique and non-replicable (implemented via secure digital token or NFT-style mechanism) to prevent scalping and fraudulent resale.

FR-06: Ticket Delivery Options

Customers shall receive tickets either digitally (via email or mobile app) or as printable/physical tickets at the box office or kiosk.

FR-07: Purchase Window

Tickets may be purchased up to 2 weeks in advance and up to 10 minutes after the scheduled showtime start.

FR-08: Optional Loyalty Accounts

Users may create optional loyalty accounts that store personal information, saved payment methods, purchase history, and accumulated loyalty points. Only one concurrent login per account is permitted.

FR-09: Discounts

The system shall support discounted pricing for students, military/veterans, and seniors (particularly on weekdays). Discount eligibility shall be verifiable at purchase time.

FR-10: Post-Purchase Feedback

After a successful purchase, the system shall prompt the customer to provide quick feedback using a simple smiley-face rating system.

FR-11: Administrative Functions

Authorized theater staff shall have access to an admin interface to set showtimes, manage theater configurations, override customer purchases when errors occur, and view reports.

FR-12: High-Demand Queueing

During periods of high demand for popular showings, the system shall implement a fair queueing mechanism to manage request volume without crashing.

FR-13: Daily Purchase Logging

The system shall maintain system-wide daily logs of all ticket purchases for auditing and reporting purposes.

FR-14: Multi-Language Support

The user interface shall be available in English, Spanish, and Swedish, with consistent pricing display (primarily USD).

3.3 Use Cases

Three primary use cases have been elaborated below following the format used in the course lecture materials (actor, flow of events, entry conditions). A use case diagram is provided at the end of this section.

UC-01: Purchase Tickets (Online)

Actor(s): Customer (guest or registered)

Purpose: Allow a customer to browse showtimes, select seats, purchase tickets, and receive confirmation for a specific movie showing.

Entry Conditions: Customer has internet access and a supported browser. The desired showing has available seats and is within the purchase window (up to 2 weeks in advance or 10 minutes after start). **Main Success Scenario (Flow of Events):**

1. Customer navigates to the SDTheaterTickets website or kiosk and browses available movies and showtimes.
2. Customer selects a specific showing (theater, date, time) and views the seat map (assigned seats for deluxe theaters).
3. Customer selects desired seats and enters the number of tickets (1–20).
4. System applies any applicable discounts (student, military, senior) if the customer provides valid credentials.

5. Customer reviews the order summary (tickets, seats, price, taxes) and proceeds to checkout.
6. Customer selects payment method (credit card, PayPal, or Bitcoin) and enters payment details.
7. System processes payment and, upon success, generates unique ticket(s).
8. Customer receives digital ticket(s) via email/app and/or prints physical ticket at kiosk.
9. System prompts customer for quick smiley-face feedback.
10. Customer logs out or continues browsing.

Extensions / Alternative Flows:

- Payment fails → System displays error and offers alternative payment method or cancellation.
- Selected seat becomes unavailable during checkout → System notifies customer and suggests closest available seats.
- 5-minute timer expires → System releases held seats and returns customer to browse.
- Single-tab Enforcement → System ensures that only one browser instance per device can be opened, displays error until a session is terminated.

UC-02: Create and Manage Loyalty Account

Actor(s): Guest Customer → Registered Customer

Purpose: Allow a customer to create an optional loyalty account to store preferences, payment methods, history, and earn points.

Entry Conditions: Customer has a valid email address and is not already logged in with another account. **Main Success Scenario:**

1. Customer selects "Create Account" or "Join Loyalty Program" from the website or kiosk.
2. Customer enters personal information (name, email, optional phone) and creates a password.
3. System validates email uniqueness and sends a verification link.
4. Customer verifies email and logs in for the first time.
5. Customer optionally adds payment methods and sets preferences (favorite theaters, genres).
6. System creates the loyalty account and awards initial welcome points.

Extensions:

- Email already registered → System offers password reset or login.
- Customer forgets password → Standard secure password reset flow via email.
- Password security → Enforces general password requirements.

UC-03: Admin Override Purchase or Manage Showtimes

Actor(s): Theater Staff / Administrator

Purpose: Allow authorized staff to correct customer purchase errors, adjust showtimes, or manage theater configurations.

Entry Conditions: Staff member has valid admin credentials and appropriate permissions. **Main Success Scenario:**

1. Staff logs into the admin portal using secure credentials.
2. Staff searches for a specific customer purchase or showing.
3. Staff selects the override action (e.g., refund, seat change, showtime adjustment).
4. System validates the change against business rules and records the action in the audit log.
5. System notifies the affected customer (if applicable) and updates all relevant displays and inventory.

Extensions:

- Attempted override violates business rules (e.g., showtime conflict) → System blocks the

action and explains the reason.

Use Case Diagram (Text Representation):

Customer — Purchase Tickets —> System

Customer — Create/Manage Loyalty Account —> System

Theater Staff — Admin Override / Manage Showtimes —> System

(<<includes>> Authenticate for protected actions)

3.5 Non-Functional Requirements

3.5.1 Performance

The system shall support a minimum of 1,000 concurrent users during peak evening hours with average page load times under 3 seconds and checkout completion under 30 seconds for typical transactions.

3.5.2 Reliability & Availability

The system shall maintain 99.5% availability during operating hours. Scheduled maintenance shall be performed during low-traffic periods with advance notice and minimal service interruption.

3.5.3 Security

All sensitive data (payment information, personal data) shall be encrypted in transit (TLS 1.3) and at rest. Tickets shall be generated as unique, non-replicable tokens. Only one concurrent session per user account is permitted.

3.5.4 Usability

The user interface shall be intuitive for users with varying technical skills. New users should be able to complete a ticket purchase with no more than 5 minutes of familiarization. The interface shall be responsive and accessible (WCAG 2.1 AA compliance target).

3.5.5 Maintainability & Portability

The system shall be built with modular, well-documented code to allow low-effort updates. Updates shall be deployable with zero or near-zero downtime. The web application shall run on standard cloud infrastructure and be portable across major cloud providers.

4. Software Design Specification

4.1 System Description

The San Diego Movie Theater Ticketing System is a web-based platform that enables customers to browse movies and showtimes across 20 theaters in the San Diego area, select and reserve specific seats (assigned seating in deluxe theaters with 75 seats; open seating in regular theaters with 150 seats), purchase up to 20 tickets per transaction, manage optional loyalty accounts, view aggregated movie reviews and critic quotes from Rotten Tomatoes and IMDb, and receive digital or physical tickets. The system supports English, Spanish, and Swedish interfaces, multiple payment methods (credit/debit cards, PayPal, Bitcoin), discounts for eligible customers, post-purchase feedback, and a 5-minute purchase timer. It includes a secure admin portal for theater staff to manage showtimes, process overrides, view reports, and handle exceptions while maintaining daily transaction logs and unique/non-replicable tickets to prevent scalping. The system is accessible via modern web browsers and in-theater kiosks, designed for high concurrency (minimum 1,000 simultaneous users during peak hours) with real-time seat availability updates.

4.2 Software Architecture Overview

4.2.1 Architectural Diagram of All Major Components

The system follows a layered architecture (Presentation → Business Logic → Data Access) with clear separation of concerns for maintainability and scalability.

Major Components and Connectors:

- Presentation Layer

- Customer Web Frontend (responsive React/Vue.js-based UI for browsing, seat selection, checkout, loyalty management)
- Kiosk Interface (simplified touchscreen UI)
- Admin Portal (staff dashboard for showtime management, overrides, reporting)

- Business Logic Layer (core API layer)

- Authentication & Authorization Module
- Ticket Booking & Reservation Module (seat map logic, 20-ticket limit, 5-min timer, single-session enforcement)
- Payment Processing Module (integrates with external gateways)
- Discount & Loyalty Module
- Review Aggregation Module (external API calls)
- Notification Module (email/SMS for tickets and feedback)

- Data Access Layer

- Database (PostgreSQL) – stores Users, LoyaltyAccounts, Movies, Theaters, Showings, Seats, Tickets, Payments, Reviews, Logs

- External Integration Layer

- Payment Gateways (Stripe/PayPal for cards & PayPal; Bitcoin API)

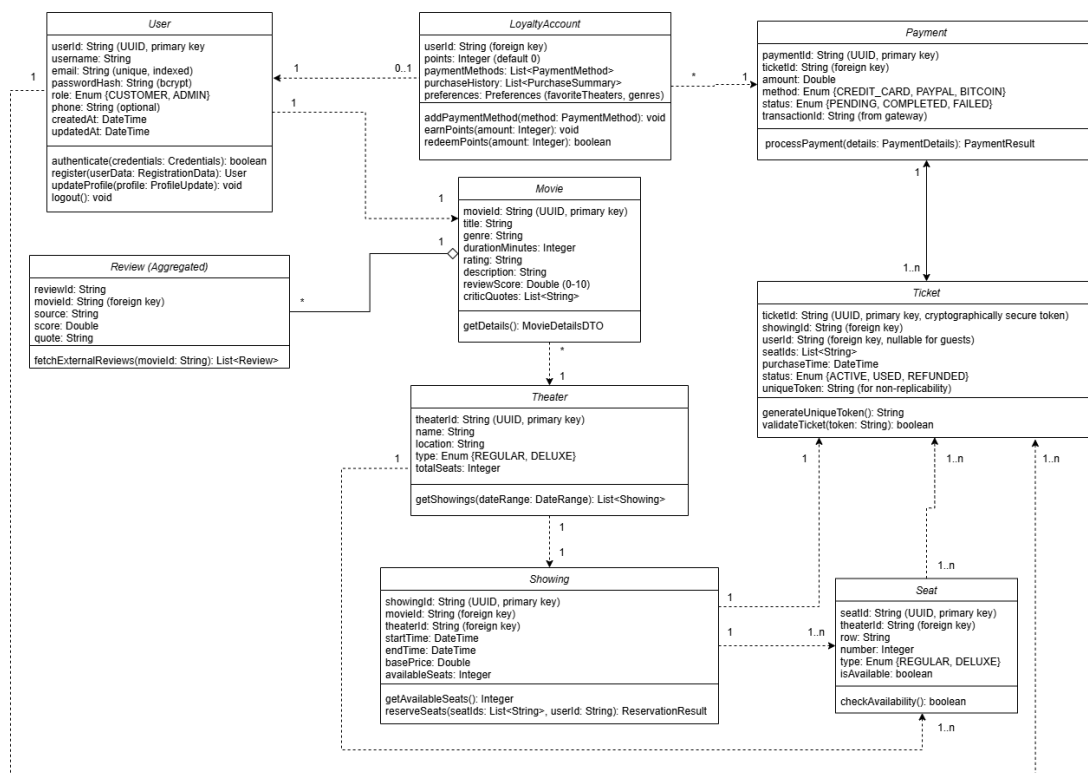
- Movie Review APIs (Rotten Tomatoes, IMDb)

- Email/SMS Gateway (e.g., SendGrid/Twilio)

Connectors/Flows:

Frontend ↔ REST/GraphQL API (HTTPS/TLS) ↔ Business Logic Modules ↔ Database (JDBC/ORM) and External Services (HTTPS). Real-time updates via WebSockets for seat availability. All sensitive data encrypted in transit (TLS 1.3) and at rest.

4.2.2 UML Class Diagram



Key Classes (with detailed attributes and operations):

Class: User

Attributes:

- userId: String (UUID, primary key)
- username: String
- email: String (unique, indexed)
- passwordHash: String (bcrypt)
- role: Enum {CUSTOMER, ADMIN}
- phone: String (optional)

- createdAt: DateTime
- updatedAt: DateTime

Operations:

- authenticate(credentials: Credentials): boolean
- register(userData: RegistrationData): User
- updateProfile(profile: ProfileUpdate): void
- logout(): void

Class: LoyaltyAccount (1:1 association with User)

Attributes:

- accountId: String (UUID, primary key)
- userId: String (foreign key)
- points: Integer (default 0)
- paymentMethods: List<PaymentMethod>
- purchaseHistory: List<PurchaseSummary>
- preferences: Preferences (favoriteTheaters, genres)

Operations:

- addPaymentMethod(method: PaymentMethod): void
- earnPoints(amount: Integer): void
- redeemPoints(amount: Integer): boolean

Class: Movie

Attributes:

- movieId: String (UUID, primary key)
- title: String
- genre: String
- durationMinutes: Integer
- rating: String
- description: String

- reviewScore: Double (0-10)
- criticQuotes: List<String>

Operations:

- getDetails(): MovieDetailsDTO

Class: Theater

Attributes:

- theaterId: String (UUID, primary key)
- name: String
- location: String
- type: Enum {REGULAR, DELUXE}
- totalSeats: Integer

Operations:

- getShowings(dateRange: DateRange): List<Showing>

Class: Showing

Attributes:

- showingId: String (UUID, primary key)
- movieId: String (foreign key)
- theaterId: String (foreign key)
- startTime: DateTime
- endTime: DateTime
- basePrice: Double
- availableSeats: Integer

Operations:

- getAvailableSeats(): Integer
- reserveSeats(seatIds: List<String>, userId: String): ReservationResult

Class: Seat

Attributes:

- seatId: String (UUID, primary key)
- theaterId: String (foreign key)
- row: String
- number: Integer
- type: Enum {REGULAR, DELUXE}
- isAvailable: boolean

Operations:

- checkAvailability(): boolean

Class: Ticket

Attributes:

- ticketId: String (UUID, primary key, cryptographically secure token)
- showingId: String (foreign key)
- userId: String (foreign key, nullable for guests)
- seatIds: List<String>
- purchaseTime: DateTime
- status: Enum {ACTIVE, USED, REFUNDED}
- uniqueToken: String (for non-replicability)

Operations:

- generateUniqueToken(): String
- validateTicket(token: String): boolean

Class: Payment

Attributes:

- paymentId: String (UUID, primary key)
- ticketId: String (foreign key)
- amount: Double

- method: Enum {CREDIT_CARD, PAYPAL, BITCOIN}
- status: Enum {PENDING, COMPLETED, FAILED}
- transactionId: String (from gateway)

Operations:

- processPayment(details: PaymentDetails): PaymentResult

Class: Review (aggregated)

Attributes:

- reviewId: String
- movieId: String (foreign key)
- source: String
- score: Double
- quote: String

Operations:

- fetchExternalReviews(movieId: String): List<Review>

Associations (text form for diagram):

User1 — 0..1LoyaltyAccount

LoyaltyAccount* — 1Payment

Payment1 — 1..nTicket

Ticket1..n — 1..nSeat

Seat1..n — 1Showing

Showing1 — 1Theater

Theater1 — *Movie

Review* — 1Movie

Movie1 — 1User

Ticket1..n — 1User

Seat1..n — 1Theater

Ticket1 — 1Showing

Movie1 — 1User

4.2.3 Description of Classes, Attributes, and Operations

(See the detailed attribute/operation lists above in 4.2.2. All datatypes are specified. Operations include clear parameter types and return types. For example, `reserveSeats(seatIds: List<String>, userId: String): ReservationResult` where `ReservationResult` is a DTO containing success flag, reserved seats, and error message if any.)

4.3 Development Plan and Timeline

Partitioning of Tasks (8-week prototype timeline)

- Week 1–2: Database design, schema implementation, initial data seeding, and Entity-Relationship modeling.
- Week 2–4: Backend API development (authentication, booking logic, payment integration, discount engine).
- Week 3–5: Frontend development (responsive UI, interactive seat maps, checkout flow, loyalty features).
- Week 4–6: Kiosk interface, external API integrations (reviews + payments), and notification service.
- Week 5–7: Admin portal, reporting/logging features, and security hardening (unique tokens, session management).
- Week 6–8: Comprehensive testing (unit, integration, end-to-end, security), bug fixing, performance optimization, and user acceptance testing.
- Week 7–8: Documentation updates, deployment to staging/production, and final polish.

Team Member Responsibilities

- Anthony Alarcon: Backend development, database design, payment integration, security features, and overall architecture. (Primary owner of Weeks 1–4 backend tasks + security.)
- Eric Phan: Frontend and kiosk UI/UX development, responsive design, seat selection interface, and customer-facing flows. (Primary owner of Weeks 3–6 frontend tasks.)
- Ven Nguyen: Admin portal, reporting & logging, external review API integration, comprehensive testing, and QA. (Primary owner of Weeks 5–8 admin/testing tasks.)

Each team member is required to make at least one commit to the shared GitHub repository. The project uses feature branches and pull requests for collaboration. Estimated completion: fully functional prototype ready for user testing by the end of Week 8.

A. Appendices

A.1 Source Requirements Traceability

All functional and non-functional requirements in this document are derived from and traceable to the client interview notes (Theater Ticketing System Qs.rtf), elicited requirements lists, and additional stakeholder input provided in July 2026.

A.2 Future Considerations (Out of Scope for v1)

- Native mobile applications
- Monthly/annual subscription passes
- Expansion beyond San Diego
- Integration with third-party ticket resale platforms
- Advanced analytics dashboard for executives

— *End of Software Requirements Specification v1.0* —